

First Name: Mahboobeh

Surname: Fattahi

Email: m.fatahi@khorasan.ac.ir

Linkedin: www.linkedin.com/in/mahboubbehfattahi

Github: github.com/mfatahi37



About me:

I am a PhD graduate in software engineering with expertise in machine learning and optimization. Seeking a post-doc position to explore innovative applications and advance impactful research in the field.

EDUCATION:

- Ph.D. Computer Engineering, Department of Computer, Islamic Azad University of Technical Engineering, Mashhad, Iran. 2018-2023.

Thesis: *"Simultaneous feature extraction and weighting by maximizing inter-class distance and minimizing intra-class distance in imbalanced classification problems"*.
Research advisor: Dr. Hossein Moattar and Dr. Yahya Forghan
- M.Sc. Computer Engineering, Department of Computer, Islamic Azad University of Technical Engineering, Ferdows, Iran. 2011-2015.

Thesis: *"Presenting a combined method to detect fraud in Financial statements and credit cards"*.
Research advisor: Dr. Hossein Moattar
- B.Sc. Computer Engineering, Computer Engineering, Department of Computer, Islamic Azad University of Technical Engineering, Mashhad, Iran. 2005-2009.

RESEARCH INTEREST:

Machine learning Algorithms (Classification and Clustering Algorithms Development)

Natural language processing

Deep learning (Convolutional neural network)

Imbalanced data classification and optimization

HONORS AND AWARDS

2018: Ranked 9th for Ph.D exam, Islamic Azad University of Technical Engineering, Mashhad, Iran.

Patented reality-based systems developed in the treatment of obsessive-compulsive fear of cockroaches.

Awarded best idea at the festival of the best idea of health.

WORK EXPERIENCES

- ❖ Khorasan Higher Education Center University- Mashhad Branch, Mashhad, Iran

Director of Graduate Studies
June 2025 – Present

- ❖ Khorasan Higher Education Center University- Mashhad Branch, Mashhad, Iran
Assistant professor
September. 2024-Persent
- ❖ Khorasan Higher Education Center University- Mashhad Branch, Mashhad, Iran
Lecturer
Oct. 2024-Persent
- ❖ Islamic Azad University- Mashhad Branch, Mashhad, Iran
Lecturer
Mar. 2020- Jul. 2024
- ❖ University Instructor of Computer Engineering, Department of Computer, Kashmir Higher Education Center, Iran
Lecturer
Dec. 2022- Jul. 2024
- ❖ Software manager at Reflection software company, Mashhad, Iran. Lecturer 2015 – 2016.
- ❖ Front-end designer at a Company published by Astan Quds Razavi, Mashhad, Iran. 2013-2014.
(<https://behnashr.ir/>)
- ❖ Researcher at Sarai Institute of Smart Technology and Health, Islamic Azad University Mashhad, Iran. 2021 up to 2022.(<http://vcr.mshdiau.ac.ir/mainsara2020.html>)
- ❖ Participated as the reviewer at the ICCKE 2021 and ICCKE 2022.
- ❖ Participated as the reviewer at the ICCKE 2023 and ICCKE 2024.
- ❖ Participated as a reviewer for a manuscript at the Asian Journal of Research in Computer Science, May 2025.

TEACHING EXPERIENCES

- ❖ University Instructor of Computer Engineering, Department of Computer, Kashmir Higher Education Center, Iran:
 - Advanced Computer Programming
 - Data Structures
 - Programming Languages
 - Logic Circuits
 - Analysis and Design of Algorithms
 - Theory of Languages and Automata
 - Web-Based Programming
 - Advanced Programming
 - Advanced Programming Workshop
 - Bachelor's Thesis Defense Judge
-
- ❖ University Instructor of Computer Engineering, Department of Computer, Khorasan Higher Education Center, Iran:
 - Computer lab
 - Computer Fundamentals and Programming
 - Advanced Programming
 - Web Design
 - Advanced Programming Workshop
 - Project Work
 - Internship
 - Bachelor's Thesis Defense Judge
 - ❖ Islamic Azad University- Mashhad Branch, Mashhad, Iran:
 - Computer lab
 - Computer Fundamentals and Programming

SKILLS

Language:

- Professional Knowledge of English.
- Familiar with Arabic
- Mother tongue: Persian

Computer Skills:

- Python (Pandas, Numpy, SKLearn, matplotlib)
- Matlab (Fuzzy, ANFIS, NN, Simulink, M-File)
- Knowledge of AI concepts
- Experience with machine learning techniques and frameworks
- Familiarity with deep learning methods and libraries
- Machine Learning Optimization
- Experience in Data Science projects using Python
- Turtle, Kivy, SQLite, Tkinter
- HTML, CSS, JavaScript
- Familiar with Django
- Experience with WordPress (including courses taken)
- Git

PUBLICATIONS (Scholar page)

1. **M. Fattahi**, M.H. Moattar. Y. Forghani, "Locally alignment based manifold learning for simultaneous feature selection and extraction in classification problems", Knowledge-Based Systems 2023.
2. **M. Fattahi**, M.H. Moattar. Y. Forghani, "Improved cost-sensitive representation of data for solving the imbalanced big data classification problem", Journal of Big Data 2022.
3. **M. Fattahi**, M.H. Moattar. "A New Hybrid Fraud Detection Approach in Credit Cards and Financial Statements", ICEEE 2015.
4. **Book:** Fraud detection in financial statements and credit cards, book (March 2017) (Persian)
5. **Book:** Data analysis models and algorithms for intelligent data analysis, book (March 2018)
6. **Book:** JavaScript Programming for Beginners and Youth, book (March 2025) (Persian)

REFERENCES

- Associate Professor Mohamad. Hossein Moattar, Department of Computer, Islamic Azad University of Technical Engineering, Mashhad, Iran.
Email: moattar@mshdiau.ac.ir
- Associate Professor Yahya Forghani, Department of Computer, Islamic Azad University of Technical Engineering, Mashhad, Iran.
Email: yforghani@mshdiau.ac.ir